

Retrieval-Augmented Detection and Explanation of Dark Patterns in E-Commerce Product Texts

Group 7

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Agenda

- Motivation & problem
 - Dataset & augmentation
 - Pipeline architecture
 - Retrieval strategies
 - Main results
- Cross-dataset & cross-lingual
 - Human evaluation
 - Limitations & conclusion
 - Demo

What is a dark pattern?

Deceptive interface and text strategies that manipulate consumers into unintended actions

"Only 2 hours left at this price!"

Urgency

"Just 3 left in stock"

Scarcity

"Cancel only by certified mail"

Forced Action

"Free trial — auto-renews at \$49.99/mo"

Sneaking

Why now?

Mathur et al. (2019) found them on 11% of 11,000 shopping sites

EU Digital Services Act (2022) mandates auditing & removal

Need: automated, explainable detection regulators can audit and operators can act on

Refs: [4], [5], [6], [7]

Existing approaches

Fine-tuned classifiers

(Yada et al. 2022 BERT)

No explanation

No evidence span

No rewrite

Zero-shot LLM (Just Prompting)

Macro F1 only 0.488

Misdirection F1 = 0.000

F1 collapses on rare classes

What we want instead

Regulators: which lever, which harm?

Operators: how to fix the snippet?

An output that can be audited

Gap: Neither option produces an output a regulator can audit or an operator can act on directly.

Refs: [1], [3]

Source corpus: EC-DarkPattern (Yada et al., 2022)

The benchmark for dark-pattern detection in e-commerce listings

What's in the corpus

- 2,361 product-description snippets
- Source: Amazon product listings
- Annotated by trained crowdworkers
- Inter-annotator agreement $\kappa > 0.7$
- 8-category taxonomy (right)

Yada et al. show fine-tuned BERT beats rule-based methods, but classification is all you get. No explanation.

The 8-category taxonomy

Category	Example signal phrase
Not Dark Pattern	—
Scarcity	"only 3 left"
Social Proof	"61 people just bought"
Urgency	"2 hours remaining"
Misdirection	"no thanks, I despise..."
Obstruction	"cancel only by mail"
Sneaking	free trial w/ auto-renewal
Forced Action	"agree to all to continue"

What we changed: re-stratified to 70/15/15 v2 splits + added 145 synthetic rare-class examples → working corpus 2,501.

Refs: [1], [3]

Synthetic augmentation: the rare-class problem

Three categories were too rare to learn or evaluate reliably

The problem

- Original test set had zero Forced Action examples
- Sneaking: 8 total; Obstruction: 27 total
- Both BERT and zero-shot LLM hit F1 = 0 here
- Zero-shot rarely sees them in context either

The method

- Composite prompting (Kochanek et al. 2024)
- Real exemplar + structured mechanism + harm
- LLM generates paraphrases of the seed
- Bigram-Jaccard filter > 0.4 to discard duplicates
- 145 examples added: 103 train, 42 val/test (stratified)

Ablation: BERT classifier macro F1 across augmentation proportions

Augmentation	Macro F1	Forced Action F1	Sneaking F1	Obstruction F1
0% (no augmentation)	0.785	0.000	0.545	0.923
33%	0.952	1.000	0.769	0.960
67%	0.964	1.000	0.933	0.923
100% (full augmentation)	0.968	1.000	0.941	0.960

Refs: [1], [2], [3]

How we generated synthetic examples

Hybrid similarity-plus-composite prompting (Kochanek et al., 2024)

Kochanek et al. (2024) — Improving Training Dataset Balance with ChatGPT Prompt Engineering.

Paper compares three prompting strategies for class-imbalance rebalancing: basic, composite, and similarity. We implement the similarity+composite hybrid the paper proposes as future work.

1. Similarity seeding

(from paper) Each call seeds the LLM with a real example from the target rare class. Keeps generated text domain-grounded rather than template fill-in.

2. Composite descriptors

Probabilistic slots assembled into the prompt:

- manipulation mechanism
- harm dimension
- product type (rotation)

3. Format & temperature

- JSON output to Python list (parse reliability).
- Temperature 0.8 (paper).
- We use Qwen3 8B locally instead of GPT-3.5-turbo.

Prompt template (simplified)

"Generate a {product_type} listing exhibiting {category} via {mechanism}. Reference example: '{seed}'. Output one JSON snippet."

Refs: [2]

Did synthetic data introduce hidden issues?

1. Bigram-Jaccard filter

Synthetic examples sharing **> 0.4** bigram overlap with any existing training example were discarded post-generation.

Prevents near-duplicate regeneration.

2. Distinct bigram ratios: 0.95 – 0.99

Between real and synthetic text.

Synthetic outputs have distinct bigram inventories, so they do not collapse to a single template.

3. Pairwise overlap

Real ↔ real bigram overlap < 0.015.
Real ↔ synthetic cross-overlap < 0.40.

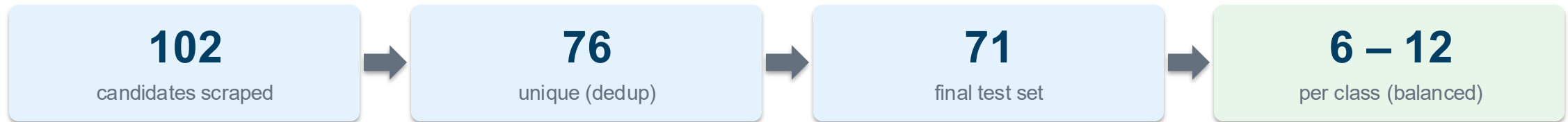
Synthetic examples are not paraphrases of specific real examples.

Honest disclosure

- 19 of 376 test examples are synthetic (5%).
- A fine-tuned classifier can implicitly memorize the synthetic generator's style.
- RAG cannot, since there are no gradient updates. Real-only re-evaluation is left for future work.

Refs: [1]

German dark-pattern test set



Sources & scope

- 4 e-commerce sites: Otto, Zalando, Shein, MediaMarkt
- Listing pages, subscription flows, checkout screens
- 3 annotators (paper authors), independent labels
- Resolved by majority vote; 5 cases without majority dropped

Annotation guidelines

Same 8-category EC-DarkPattern taxonomy, augmented with German signal phrases:

"Nur noch X Stück" → Scarcity

"automatische Verlängerung" → Sneaking

"Kündigung nur per Einschreiben" → Obstruction

Refs: [1]

System architecture: Two-Phase Pipeline

PHASE 1: Offline

Build the searchable library

- ① Take 1,750 labelled training texts
- ② Tokenize each text → token IDs
- ③ Forward pass-through encoder
- ④ Mean-pool → single 768-dim vector
- ⑤ L2-normalise the vector (length = 1)
- ⑥ Store in FAISS flat index



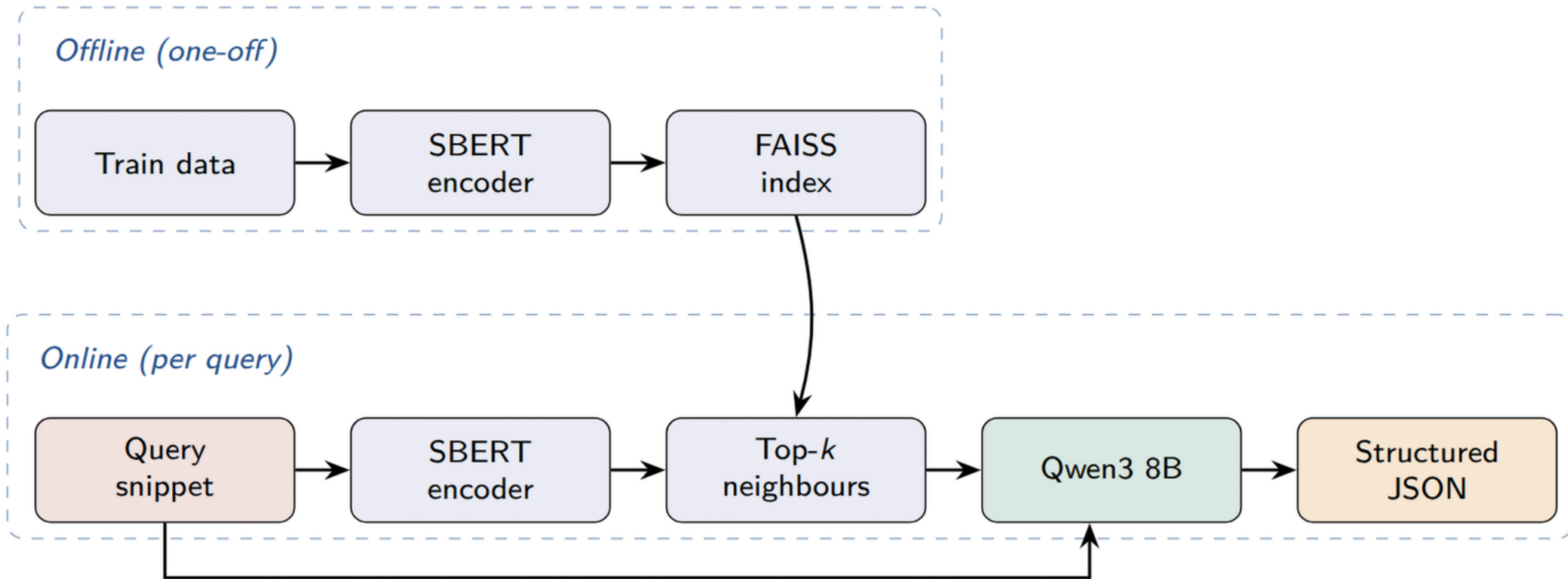
PHASE 2: Online (every new text)

Retrieve, prompt, generate

- ① New suspicious text arrives
- ② Encode with same SBERT encoder
- ③ Search FAISS → top-k similar examples
- ④ Build prompt: k examples + new text
- ⑤ Qwen3 8B generates structured JSON
- ⑥ Return 7-field output to user

Refs: [1],[11],[16],[17]

System architecture: Two-Phase Pipeline



Refs: [1],[11],[16],[17]

Encoders tested in Phase 1

Encoder	Pooling	Dim	Params	F1	Used for	Why it matters
SBERT	MEAN	768	110M	0.923	English	- NLI fine-tuned - mean-pool gives rich semantic vectors. Best for similarity.
BERT	CLS	768	110M	0.765	English	Same size as SBERT, but CLS pooling was trained for Masked Language Models, not similarity.
RoBERTa-Large	CLS	1024	355M	0.773	English	3× larger than BERT but still CLS pooling

Refs: [1], [11], [12], [13]

Retrieval strategies compared

KNN: K-nearest neighbours

- Picks the k most similar training vectors using cosine similarity
- It works best because dark patterns carry strong surface signals, i.e., the nearest example almost always shares the correct label.

Prototype

- Scores each candidate by its similarity to both the query AND the center (prototype) of its class
- More representative picks, but adds noise for clear-signal categories.

MMR / Diversity

- Maximal Marginal Relevance iteratively picks examples that are relevant to the query but diverse from already selected ones.
- Prevents identical-class slots, but dark patterns don't need diversity.

HyDE: Hypothetical document embeddings

- LLM first imagines an ideal dark-pattern snippet (temp=0.7), encodes that hypothetical text, then retrieves neighbours using it
- Adds noise when original queries are already short and clear.

Refs: [1], [9], [10]

Structured output schema: 7 fields per prediction

label

The dark pattern category is one of 8 types (e.g., urgency, scarcity, forced action).

confidence

- Model's self-rated certainty (0–1)
- Low-confidence cases can be flagged for human review by regulators.

psychological_mechanism

The cognitive bias being exploited (e.g., loss aversion, social conformity, status quo bias)

evidence_span

- Verbatim substring from the input text that is manipulative
- Lets regulators point to the exact phrase.

harm_dimension

The consumer harm (e.g., financial loss, privacy violation, autonomy deprivation)

rationale

- 1–3 sentence explanation connecting the evidence span → mechanism → harm
- Human-readable.

rewrite

- A non-manipulative alternative that keeps product info but removes the deceptive element
- Actionable for platforms.

Refs: [1]

Experimental setup & comparison design

Controlled local inference, fixed prompt/schema, and targeted ablations

MODEL

Qwen3 8B

Frozen, locally served via Ollama

GENERATION

Temperature = 0.0

JSON output with fallback parser

RETRIEVAL

FAISS flat

Exact inner-product search

HARDWARE

RTX 5060

8 GB VRAM, 32 GB RAM

Tier	Question tested	Configurations
No retrieval	What happens without grounding examples?	Zero-shot; random few-shot k=3; random few-shot k=5
RAG with SBERT	Which retrieval strategy works best?	KNN k=5; KNN k=2; Diversity/MMR; Prototype; HyDE
Alternative encoders	Does representation quality matter?	BERT KNN k=5; RoBERTa KNN k=5; RoBERTa KNN k=2

Baselines and evaluation metrics

How configurations are compared and what is measured

Supervised baseline: Fine-tuned BERT

bert-base-uncased · fine-tuned for classification only

Training	5 epochs, lr = 2e-5, batch 16, AdamW
Train data	1,647 real examples
Output	Label only, no explanation, no rewrite
Purpose	Baseline performance for our RAG pipeline
Result	Macro F1 = 0.785

Classification

Macro-averaged F1, precision, recall · Per-class F1 across all 8 dark-pattern categories · Cohen's κ (chance-corrected agreement with gold labels)

Span grounding

Exact-match rate of predicted evidence_span · Case-insensitive exact match rate · Mean span length in characters

Confidence calibration

Calibration gap = |mean confidence – accuracy| · Smaller gap = model knows when it is uncertain · Compared across all 11 configurations

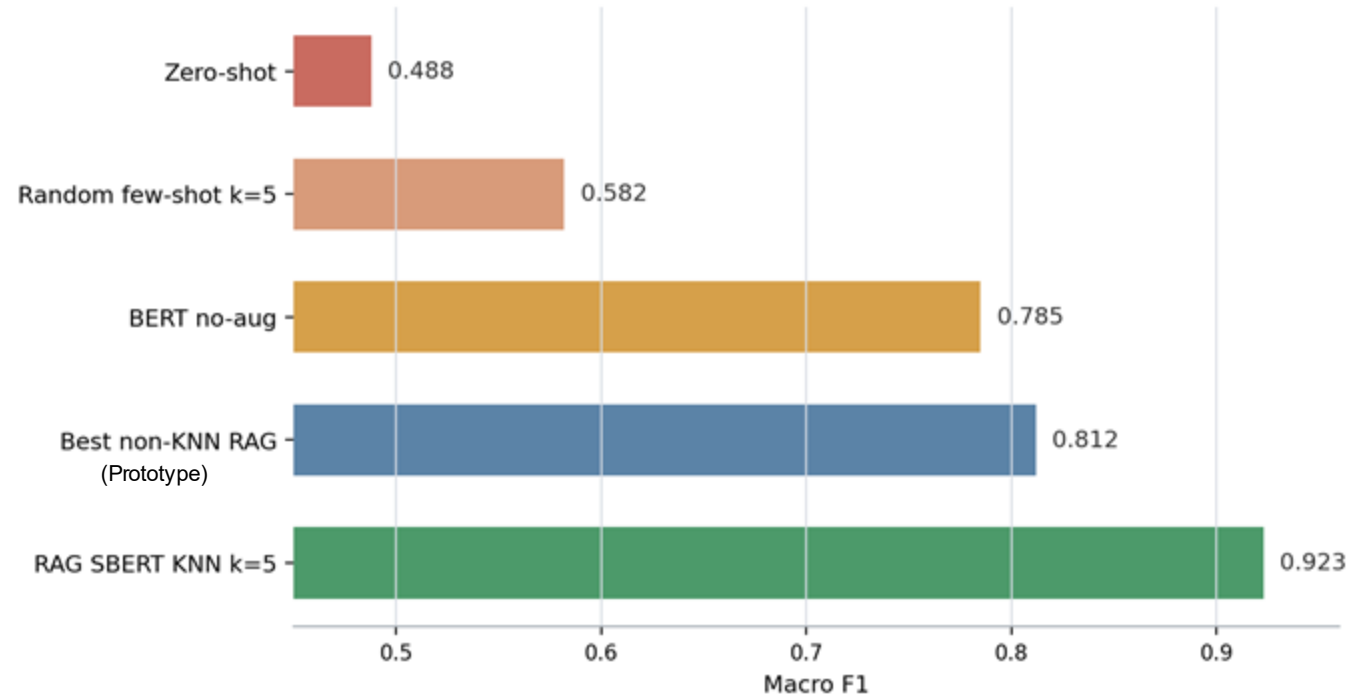
Cross-lingual gap

Absolute F1 drop from English to German test set · Per-class transfer behavior inspected · English-only vs multilingual encoder compared

Refs: [1], [13], [15]

Main result: retrieval relevance drives performance

The gain comes from picking relevant demonstrations, not from adding more examples



+0.435

vs zero-shot

+0.341

vs random few-shot k=5

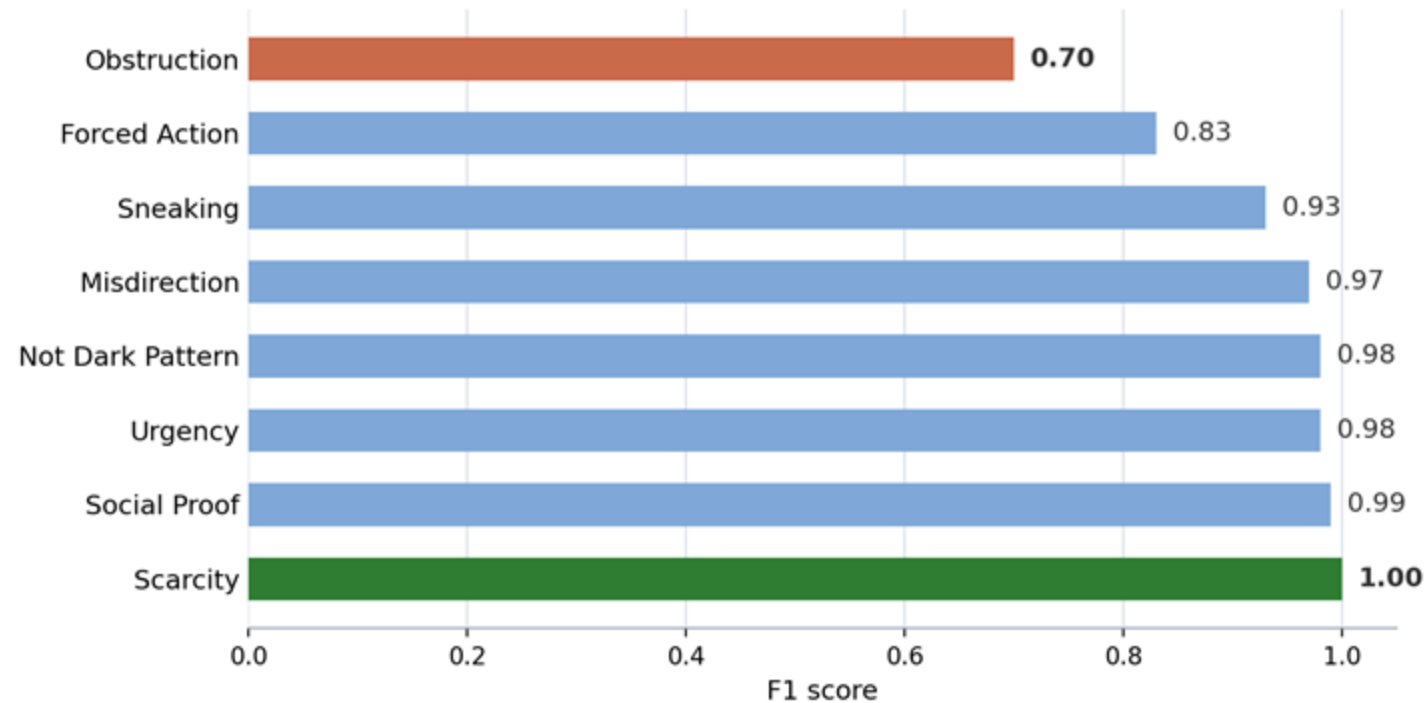
+0.138

vs BERT no-augmentation

Same context budget, very different results. What matters is which labelled examples you retrieve, not how many.

Per-class F1 on EC-DarkPattern v2

Deployed English configuration: RAG SBERT KNN k=5



0.923

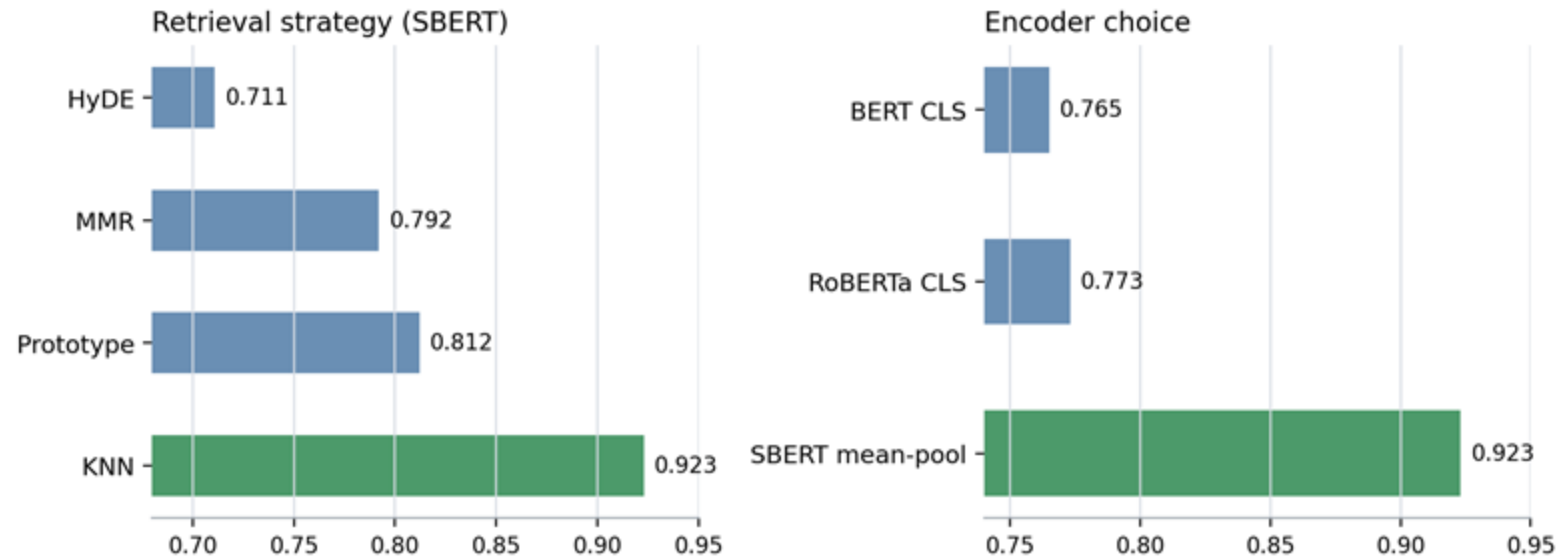
macro F1 on EC-DarkPattern

Weakest class: Obstruction (0.70).
Signal-rich classes such as Scarcity
and Urgency are near-perfect.

Overall pattern:
Most classes are well above 0.9.
The remaining difficulty is
concentrated in the structural
categories like Obstruction.

Why RAG works

Nearest labelled examples anchor the category; SBERT gives the best retrieval representation



Finding 1

KNN beats Diversity/MMR, Prototype, and HyDE because dark-pattern snippets often contain direct surface cues.

Finding 2

SBERT mean-pooling beats larger CLS-pooled encoders. Representation quality matters more than raw model size for retrieval.

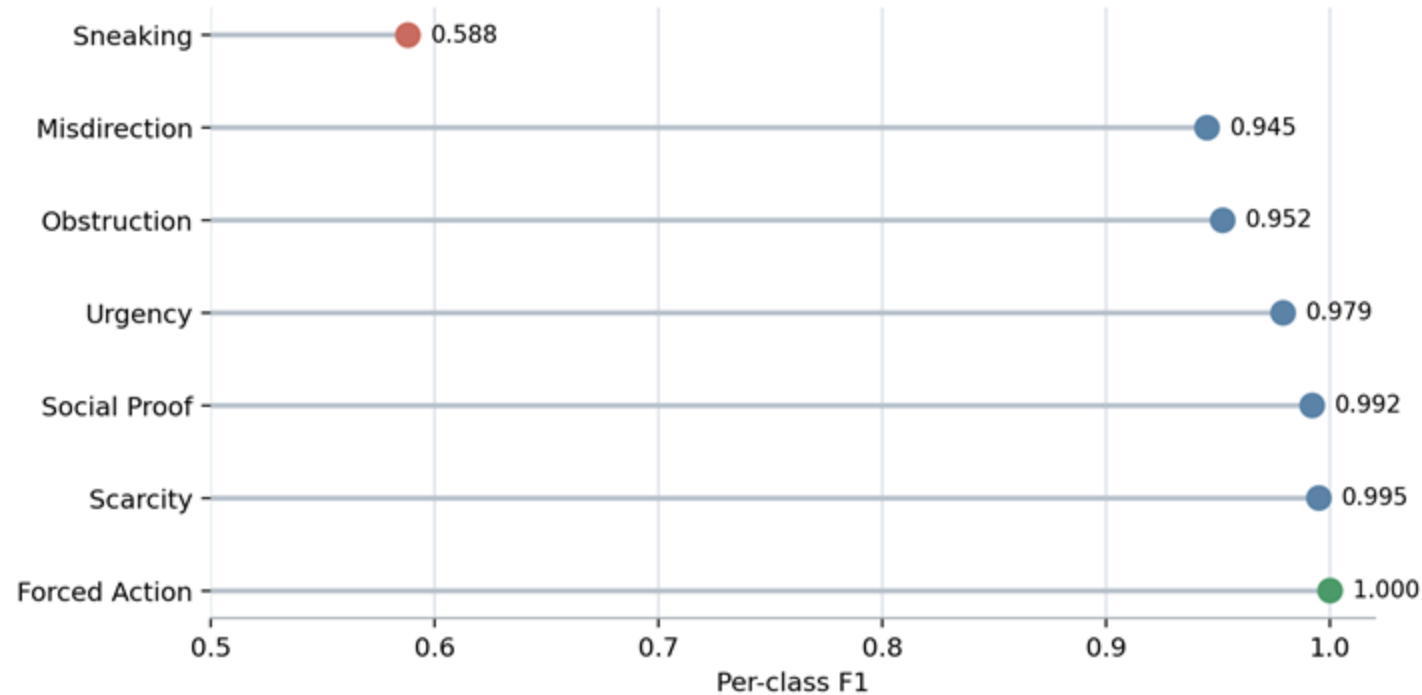
Takeaway: for this task, the right inductive bias is nearest examples plus sentence-level embeddings.

We choose RAG over BERT and fine-tuning as RAG is more flexible and meets the requirements such as to provide rationale and rewrites reliably.

Refs: [1], [11]

Cross-dataset transfer: Mathur et al. (2019)

The deployed index transfers to a related English corpus without retraining or index updates



0.922

macro F1 on Mathur

6 of 7 shared categories are at or above 0.94.

Sneaking is the weak point because annotation conventions differ.

Within-domain transfer:
both corpora are English e-commerce text.

Refs: [1], [4]

Cross-lingual evaluation: English training, German testing

We compare zero-shot German, English-encoder retrieval, and multilingual retrieval on 71 majority-labelled examples

Zero-shot German

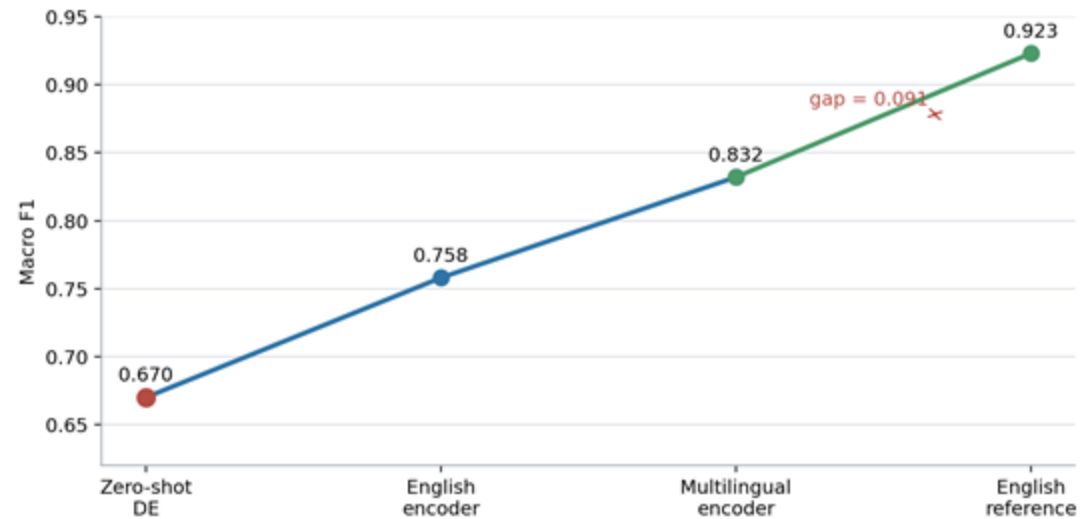
No retrieved examples. German signal phrases in the prompt only.

English-encoder retrieval

German queries retrieve from the English FAISS index, but language mismatch remains.

Multilingual retrieval

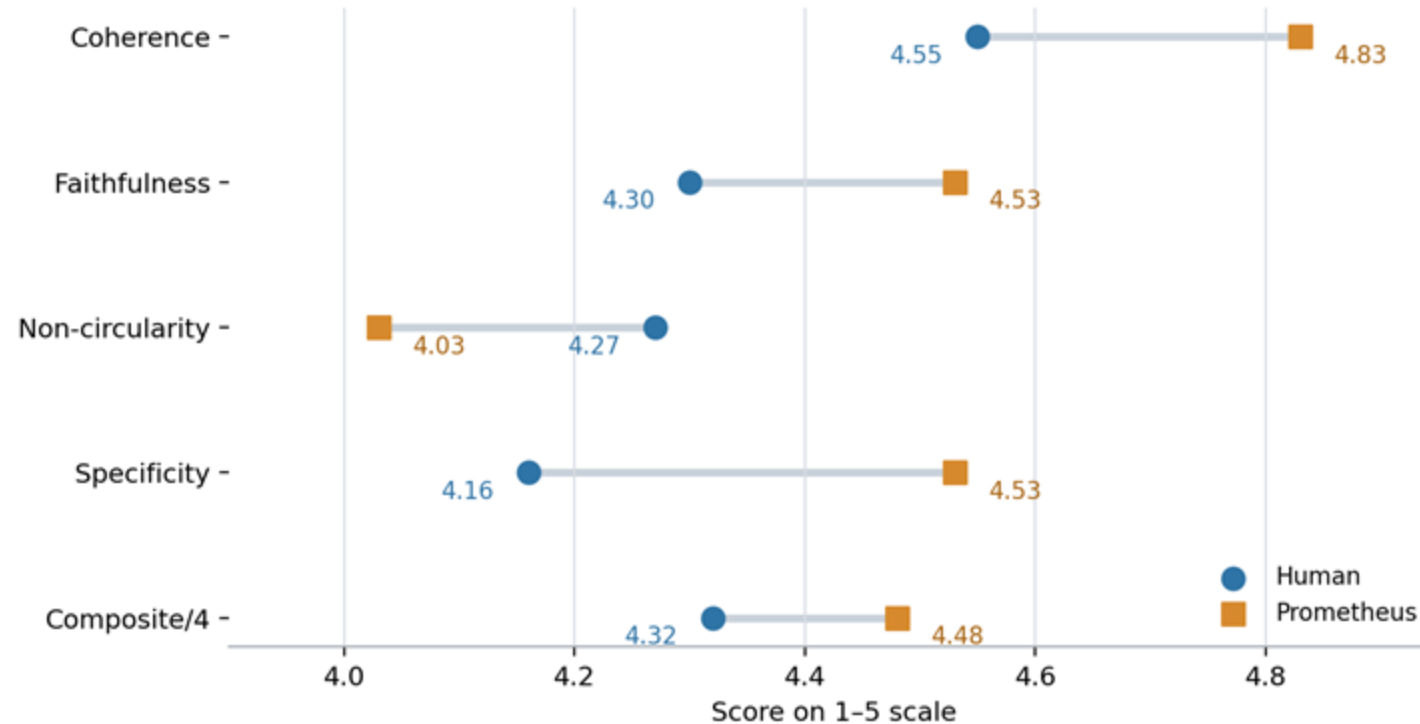
German queries and English examples are mapped into a shared multilingual space.



Refs: [1], [12]

Evaluation: explanations are strong, but agreement is weak

High mean scores support usefulness, but weak agreement limits how strongly we can validate explanation quality



17.29

human composite / 20

17.93

Prometheus / 20

Within-one agreement stays relatively high at 70.0–85.8%, but chance-corrected agreement is weak. Frame as a feasibility signal, not gold-standard validation.

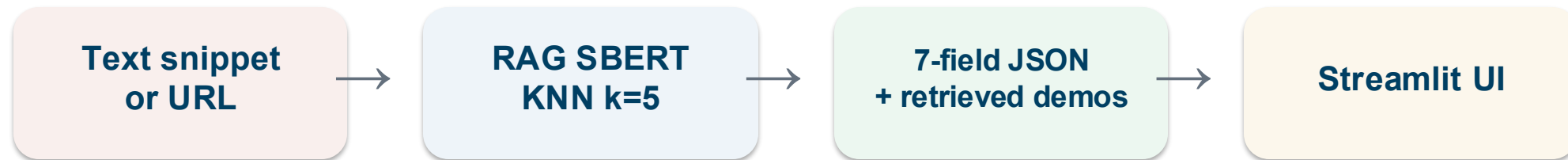
Refs: [1], [14]

Limitations

The claims are scaled to the evidence, not stretched beyond it

Limitation	What it affects
Synthetic examples in test set	Rare-class F1 may be inflated
German artifact completeness	Cross-lingual reproducibility is weaker than English
Confounded multilingual condition	Cannot isolate encoder, threshold, and MMR effects
Human evaluation scope	Small sample and weak agreement
Detecting dark pattern scripts	Dark patterns implemented in JavaScript
Single-label assumption	Multi-pattern snippets are not handled

Demo: from snippet to regulator-facing output



`streamlit run demo/streamlit_app.py`

The UI highlights the evidence span, colour-codes the label, and shows a consumer-safe rewrite for review.

References

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- [5] European Commission (2022). Digital Services Act, Regulation (EU) 2022/2065.
- [6] Brignull (2010). Dark Patterns: Deception vs. Honesty in UI Design.
- [7] Gray et al. (2018). The Dark (Patterns) Side of UX Design.
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- [9] Carbonell and Goldstein (1998). The Use of MMR, Diversity-Based Reranking for Reordering Documents and Producing Summaries.
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- [12] Reimers and Gurevych (2020). Making Monolingual Sentence Embeddings Multilingual using Knowledge Distillation.
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- [14] Kim et al. (2024). Prometheus: Inducing Fine-Grained Evaluation Capability in Language Models.
- [15] Cohen (1968). Weighted Kappa.
- [16] Johnson, Douze, Jégou (2021). Billion-Scale Similarity Search with GPUs.
- [17] Qwen Team (2023). Qwen Technical Report.

Takeaways

Explainable output

Label, confidence, evidence span, mechanism, harm, rationale, rewrite

Strong transfer

- 0.923 on EC-DarkPattern dataset
- 0.922 on Cross dataset
- 0.832 on German test data

No task-specific fine-tuning

Frozen Qwen3 8B plus dense retrieval gives a practical local system

Thank you

Questions?

Group 7: Jobin Roy · Akshita Sakshi · Shifin Mohammed

HCNLP 2026, Otto von Guericke University Magdeburg